

DataSensei Updates 3 — Change Report

Admin-editable public modules, challenges and coding challenges; instructor challenge builder; two fixes. Prepared 25 September 2026 against the codebase you uploaded, with every ZIP delivered since applied on top.

What was asked, and what was done

Request in the PDF	Delivered
After the last lesson of a module, go back to the module page instead of challenges	Done. Completing the final lesson now returns to /module with the "Module Completed!" message. Earlier lessons still go to the next lesson.
Remove the level icons (Newbie, Intermediate, ...) on /challenges and coding challenges	Done on both pages. The card title now takes that space; nothing else on the cards changed.
All predefined modules at /module editable by the admin: add block of code, text (fonts, sizes, bold, italic, inline code), table, photo — with a live web preview	Done. Module Manager (create, edit, reorder, delete, year level, XP, boss flag, coding-exercises checkbox) and Module Content Manager per module: a block editor with Code, Text, Table, Photo and HTML blocks, a live preview that is the learning room's own rendering, and photo upload.
Existing modules stay exactly as they are, but become editable	Done, as you chose. Each of the 56 existing lessons opens as one "Original lesson HTML" block, byte for byte. Saving it unchanged leaves the stored content identical (tested with CRLF, unicode and script content). New blocks can be added above or below it.
Challenge maps editable (paths and contents); Challenge Manager / Module Content Manager	Done. Challenge Maps edits each level's name, description, audience and order, and lists every MCQ and coding challenge per level with reorder, availability toggle and links to their editors.
New module automatically creates challenges, locked/unavailable; in every level; coding one only when the module has coding exercises	Done, as you chose. A new module creates one unavailable MCQ challenge in every level, and one unavailable coding challenge in every level when "has coding exercises" is ticked. Verified: 5 levels → 10 challenges, all unavailable, all with unique codes.
Challenges strictly multiple choice, with pictures, with live preview	Done. The MCQ editor gained a picture per question (uploaded, validated, shown to students on the quiz) and a live preview panel. It stays multiple-choice only.
Coding challenges customizable the CodeChum way, with test cases	Done. Coding Challenges manager: problems with title, description, starter code, reference solution, per-problem time limit and XP, and test cases (stdin, expected output, hidden). "Check test cases" runs the reference solution through the real sandbox against every case, with the exact grading rule students face. Live preview of what learners will see.
Instructors customize assignments like the admin's challenges, but cannot create modules; assignments show admin challenges and the instructor's own	Done, as you chose. Challenge Builder lets an instructor build their own MCQ (with pictures) and coding challenges (with test cases, "check test cases"); Class Challenges gives them, or an admin pool challenge, to a class with a status and dates. Students see them under Assignments ("Challenges from your instructor") and take them with the existing quiz and coding screens. Instructors cannot create modules.
Report of the changes via PDF	This document.

Decisions you answered, applied exactly

Existing lessons kept as one editable block (not split). Instructors build their own challenges with test cases. A new module fans out to every level; coding challenges only when the checkbox is ticked. One delivery.

Screens

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Module Manager



The public curriculum students work through in order. Reorder modules, edit their details, or open one to manage its lessons.

Modules

27 modules across 4 year levels. Students unlock them in the order shown here. A new module also gets one locked challenge on each of the 5 challenge levels.

Add module

Year 1

Order	Module	Lessons	XP	Type	Coding exercises	Actions
1	Basics of Python Programming Master the core concepts of Basics of Python Programming through rigorous interactive modules.	11	1000	Standard	No	↓ Lessons Edit
2	Basics of Statistics Master the core concepts of Basics of Statistics through rigorous interactive modules.	11	1000	Standard	No	↑ ↓ Lessons Edit
3	Introduction to Data Science Master the core concepts of Introduction to Data Science through rigorous interactive modules.	12	1000	Standard	No	↑ ↓ Lessons Edit
4	Mathematical Analysis I Master the core concepts of Mathematical Analysis I through rigorous interactive modules.	0	1000	Standard	No	↑ ↓ Lessons Edit
5	Methods of Proof Master the core concepts of Methods of Proof through rigorous interactive modules.	0	1000	Standard	No	↑ ↓ Lessons Edit
6	Modeling and Simulation Master the core concepts of Modeling and Simulation through rigorous interactive modules.	0	2500	Boss module	No	↑ ↓ Lessons Edit

Year 2

Order	Module	Lessons	XP	Type	Coding exercises	Actions
7	Algorithms & Data Structures for Data Scientists Master the core concepts of Algorithms & Data Structures for Data Scientists through rigorous interactive modules.	0	1000	Standard	No	↑ ↓ Lessons Edit

Module Manager: the public curriculum grouped by year, with lesson counts, coding-exercises flag, reorder, and links to each module's lessons.

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Edit Lesson

Editing a lesson in Basics of Python Programming. Build the lesson from blocks on the left; the preview on the right shows it exactly as students will read it.

Lesson

The title appears in the module's lesson list and in the learning room outline.

Save lessonCancel

Title

1.1 Foundations: Syntax, Output, Comments

Blocks

Add blocks in reading order. Code windows carry the same "Try in Compiler" button as existing lessons.

Add block

Code

Text

Table

Photo

HTML

1. HTML

Move upMove downDelete

HTML

<h2>Python Foundations</h2>

<p>Python is an interpreted, high-level, general-purpose programming language created by Guido van Rossum in 1991. It is the dominant language in data science because of its clean syntax, massive ecosystem of libraries, and ease of prototyping. Unlike compiled languages such as C++ or Java, Python executes line-by-line through an interpreter — which means you can test code instantly without a build step.</p>

<h3>Why Python for Data Science?</h3>

<p>Python dominates the data science world because of

Raw HTML, shown as is. Use this only for markup the other blocks cannot express.

2. Text

Move upMove downDelete

Font

Size

Live preview

Rendered with the learning room's styles. It refreshes as you type.

Python Foundations

Python is an **interpreted, high-level, general-purpose** programming language created by Guido van Rossum in 1991. It is the dominant language in data science because of its clean syntax, massive ecosystem of libraries, and ease of prototyping. Unlike compiled languages such as C++ or Java, Python executes line-by-line through an *interpreter* — which means you can test code instantly without a build step.

Why Python for Data Science?

Python dominates the data science world because of three reasons: readability (code looks like pseudocode), ecosystem (NumPy, Pandas, Scikit-Learn, TensorFlow all use Python as their primary API), and community (the largest open-source data science community on the planet). When you write Python, you spend more time thinking about your data — not fighting your language.

Your First Script: The `print()` Function

The `print()` function outputs data to the standard output device — your terminal or console. You will use this constantly for debugging.

Module Content Manager: an existing lesson opened as its original HTML block on the left; the live preview on the right is the learning room's own rendering, including the "Try in Compiler" buttons.

Create a versioned coding challenge with one or more problems and their test cases. New challenges are unavailable until published.

Challenge Identity and Version

The content code groups versions of the same challenge. Only one version per content code is available to learners at a time.

Difficulty Category

Newbie

Content Code

Leave blank to generate on

Version Number

1

Version Name

Version 1

Version Code

V1

Availability

Unavailable / Draft

Challenge Settings

The title and description learners see on the map, plus the challenge-level timer, XP and map order.

Title

E2E Sum Two Numbers

Time Limit in Seconds

1800

Base XP

100

Map Order

0

Description

Problems

Learners solve the problems in order. Each problem is graded against its test cases: a case passes when the program's output matches the expected output exactly, ignoring trailing whitespace.

Add Problem

Coding Challenges: a problem with reference solution and test cases; "Check test cases" ran the solution in the sandbox and reported the case as passed. The right-hand panel previews what learners see.

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Challenge Maps

Every challenge level with its MCQ and coding challenges. Edit the level text, set its position, and arrange the challenges learners meet on each map.

Newbie

Slug newbie (fixed, used in learner URLs). Position 1. 4 MCQ, 5 coding.

Level Name

Newbie

Target Audience

Beginners

Position

1

Description

Introductory questions for first-time learners building basic confidence.

Save Level

MCQ Challenges

Create MCQ Challenge

Order	Challenge	Module	Version	Questions	Status	Actions
1	Curl Module C1-MCQ-CURL-MODULE-6A40EB3A	Curl Module	Version 1 (V1)	0	Unavailable	Edit Make available
3	E2E Module With Coding C1-MCQ-E2E-MODULE-WITH-CODING-779000E4	E2E Module With Coding	Version 1 (V1)	0	Unavailable	Edit Make available
5	E2E Module With Coding M29-NEWBIE-MCQ	E2E Module With Coding	Version 1 (V1)	0	Unavailable	Edit Make available
7	E2E Module With Coding M30-NEWBIE-MCQ	E2E Module With Coding	Version 1 (V1)	0	Unavailable	Edit Make available

Coding Challenges

Create Coding Challenge

Challenge Maps: each level editable inline, with its MCQ and coding challenges, availability and order.

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Class Analytics

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ILO Mastery

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Prof. Demo Instructor

Quiz challenge "E2E Instructor Quiz" saved. Give it to a class from Class Challenges so students can take it.

Challenge Builder

Quiz and coding challenges you built. They live on the University Student level and reach students only through the classes you give them to.

Give a challenge to a class Browse the platform challenge pool

New quiz challengeNew coding challenge

Title	Type	Items	Availability	Given to	Actions
E2E Instructor Quiz	Quiz	1	Available	No class yet	<button>Edit</button> <button>Delete</button>
E2E Instructor Quiz	Quiz	1	Available	No class yet	<button>Edit</button> <button>Delete</button>
E2E Instructor Quiz	Quiz	1	Available	1 class	<button>Edit</button> <button>Delete</button>

Instructor Challenge Builder: the instructor's own challenges, with the admin pool alongside.

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Class Challenges

Give one of your own challenges, or an available University Student challenge from the pool, to a class. Students see it on the University Student challenge map and on their assignments page while it is published and inside its window.

Build your own challenges Browse the platform challenge pool

Give a challenge to a class

Each challenge can be given to a class once. Leave the dates empty to keep it open for as long as it is published.

Class

Challenge

Status

Choose class

E2E Instructor Quiz (Quiz)

Published

Title shown to students (optional) defaults to the title

Available from (optional)

Defaults to the challenge title

mm/dd/yyyy, --:-- --

Instructions (optional)

Due (optional)

Anything the class should know before starting

mm/dd/yyyy, --:-- --

After the due date the challenge disappears from the class again.

Give to class

Data Science Defense Class, DEMO-A

1 challenge given to this class

Challenge	Type	Status	Available from	Due	Students	Actions
E2E Instructor Quiz E2E Instructor Quiz (yours)	Quiz	Published	Immediately	No due date	Open now	<button>Remove</button>

Edit status, dates, title or instructions

Class Challenges: giving a challenge to a class with status and dates.



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Data Science Learning Platform



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My Assignments

View activities from your classes, continue unfinished work, and open completed results.

Status

All Statuses

Filter

Assignment	Class	Type	Due	Status	Action
Python Foundations — Timed Demo Python Foundations, Defense Version	Data Science Defense Class	MCQ	Sep 28, 2026, 09:29 AM	Not Started	Start

Challenges from your instructor

Quiz and coding challenges your instructor gave to your class. They open on the University Student challenge map.

Challenge	Class	Type	Due	Action
E2E Instructor Quiz	Data Science Defense Class	Quiz	No due date	Start quiz

Student Assignments: the new "Challenges from your instructor" section.



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Data Science Learning Platform



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Challenges

Choose a path that matches your current experience level.

Paths now unlock through progression. Complete the previous difficulty to move forward; exceptional speed and accuracy can unlock the next-next difficulty early.

Newbie
Beginners

Introductory questions for first-time learners building basic confidence.

Select Path →

University Student
College learners

Foundational academic exercises aligned with early programming and data science coursework.

Select Path →

Intermediate
Developing practitioners

Applied problem-solving challenges for learners who already understand the basics.

Complete University Student to unlock Intermediate. Exceptional performance in Newbie can unlock it early.

Locked 🔒

Advanced
Advanced learners

Higher-level analytical and coding challenges for strong performers.

Complete Intermediate to unlock Advanced. Exceptional performance in University Student can unlock it early.

Locked 🔒

Professional
Career-ready learners

Professional-level tasks focused on readiness for data-driven technical work.

Complete Advanced to unlock Professional. Exceptional performance in Intermediate can unlock it early.

Locked 🔒

Student /challenges: level cards without the icons.

How it fits the existing system

Lesson content. A new column `lessons.blocks` holds the editor's blocks as JSON text. `lessons.content` stays the rendered HTML the learning room already shows, so the student side needed no change. One service, `LessonBlockRenderer`, produces that HTML for both the live preview and the saved lesson, so what the admin sees while editing is exactly what students get. New code blocks reproduce the hand-written lessons' markup, including the colour spans and the "Try in Compiler" button, so old and new blocks look the same.

Challenges. Fanned-out challenges carry `challenges.module_id`. Instructor-built ones live in the existing *University Student* level (already unlocked by class enrolment) with `visibility = instructor` and `created_by`, and are shown only to students whose class has an open `class_challenge_assignments` row. They do not count toward level completion. A direct URL without such a row is a 404.

Coding challenges use the existing `coding_questions` and `test_cases` tables that the student grader already reads, plus two new columns (`title`, `reference_solution`). "Check test cases" uses the same comparison as grading: trailing whitespace ignored, otherwise exact.

Images are stored under `public/uploads/lessons/` and `public/uploads/challenges/` (no `storage:link` needed on XAMPP). Uploads are validated as images up to 5 MB; only those paths are accepted in content.

Database (MySQL 5.5)

One additive migration, guarded with `information_schema` checks like the existing ones: no JSON columns, no foreign keys, no generated columns, nothing that touches existing rows. It adds `modules.has_coding_exercises`, `lessons.blocks`, `challenges.module_id / created_by / visibility`, `challenge_questions.image_path`, `coding_questions.title / reference_solution`, and the table `class_challenge_assignments`. Running it twice is a no-op (verified).

Two things found and fixed during verification

Both admin managers required a typed "content code" whose placeholder made it look optional; creating a coding challenge stalled on it. The field is now optional and a unique code is generated when it is blank. Separately, the model derived codes from the title alone, so two challenges with the same title on one level, or two modules with the same title, would have collided on a unique index. Generated codes now carry a unique tail, and fanned-out challenges use an explicit code per module and level. Both are covered by tests.

Verification

Check	Result
PHPUnit, full suite	1,314 passed, 8,618 assertions, 30 skipped (the Docker-only sandbox tests)
New tests for this delivery	99 tests across 13 files (Updates3*Test.php, LessonBlockRendererTest.php), each proving authorization, validation and persistence
Python sandbox suite	72 passed, 1 skipped
Every Blade template compiles	Passed (whole-tree sweep)
Migration on MariaDB, ONLY_FULL_GROUP_BY	Applied; every column and the new table present; second run is a no-op
Every page as every role on the migrated database	385 parameterless + 190 parameterised renders across student, instructor, admin, institution admin, superadmin: no server errors
Real browser (Chromium) drive of the new screens	20 of 20 checks passed: module create with fan-out (10 challenges), legacy lesson preview, live text block, photo upload in preview, lesson save, MCQ live preview, coding check-tests (real sandbox: passed), coding save, challenge maps, instructor build, class assignment, student sees it on Assignments and on the map, level icons gone, module end returns to /module with the flash
Student side after editing	Learning room shows original content plus the new heading and photo, with working "Try in Compiler" buttons; an untouched lesson's stored HTML is byte-identical

Not verified here

A browser on Windows itself (the drive above ran on Linux against the same code), and the coding check with your Docker runner (it ran through the local driver here; the call is identical). Everything else in this report was executed, not assumed.

Files in the ZIP

Only files changed or added since your last delivery; folder structure preserved. Unzip over C:\xampp\htdocs\ so the DataSensei folder merges.

Path	Change
app/Http/Controllers/AdminChallengeMapController.php	added
app/Http/Controllers/AdminCodingChallengeController.php	added
app/Http/Controllers/AdminLessonController.php	added
app/Http/Controllers/AdminMcqChallengeController.php	modified
app/Http/Controllers/AdminPublicModuleController.php	added
app/Http/Controllers/ChallengesController.php	modified
app/Http/Controllers/CodingQuizController.php	modified
app/Http/Controllers/InstructorChallengeBuilderController.php	added
app/Http/Controllers/InstructorClassChallengeController.php	added
app/Http/Controllers/LessonController.php	modified
app/Http/Controllers/StudentAssignmentController.php	modified
app/Models/Challenge.php	modified
app/Models/ChallengeQuestion.php	modified
app/Models/ClassChallengeAssignment.php	added
app/Models/CodingQuestion.php	modified
app/Models/Lesson.php	modified
app/Models/Module.php	modified
app/Services/ChallengePathUnlockService.php	modified
app/Services/CodingChallengeTestRunner.php	added
app/Services/LessonBlockRenderer.php	added
app/Services/PlatformContentService.php	modified
bootstrap/app.php	modified
database/migrations/2026_09_24_000001_add_content_management_to_modules_and_challenges.php	added
public/js/admin-coding-challenge-editor.js	added
public/js/admin-content-manager.js	modified
public/js/admin-lesson-editor.js	added
public/js/admin-mcq-preview.js	added
public/js/instructor-challenge-builder.js	added
resources/views/admin/challenge-maps/index.blade.php	added
resources/views/admin/challenges/_form.blade.php	modified
resources/views/admin/challenges/_preview.blade.php	added
resources/views/admin/challenges/show.blade.php	modified
resources/views/admin/coding-challenges/_form.blade.php	added

Path	Change
resources/views/admin/coding-challenges/_preview.blade.php	added
resources/views/admin/coding-challenges/create.blade.php	added
resources/views/admin/coding-challenges/edit.blade.php	added
resources/views/admin/coding-challenges/index.blade.php	added
resources/views/admin/coding-challenges/show.blade.php	added
resources/views/admin/modules/_form.blade.php	added
resources/views/admin/modules/create.blade.php	added
resources/views/admin/modules/edit.blade.php	added
resources/views/admin/modules/index.blade.php	added
resources/views/admin/modules/lessons/_editor.blade.php	added
resources/views/admin/modules/lessons/_preview_frame.blade.php	added
resources/views/admin/modules/lessons/edit.blade.php	added
resources/views/admin/modules/lessons/index.blade.php	added
resources/views/instructor/challenge-builder/_coding_form.blade.php	added
resources/views/instructor/challenge-builder/_editor.blade.php	added
resources/views/instructor/challenge-builder/_mcq_form.blade.php	added
resources/views/instructor/challenge-builder/_styles.blade.php	added
resources/views/instructor/challenge-builder/create.blade.php	added
resources/views/instructor/challenge-builder/edit.blade.php	added
resources/views/instructor/challenge-builder/index.blade.php	added
resources/views/instructor/class-challenges/index.blade.php	added
resources/views/partials/admin-sidebar.blade.php	modified
resources/views/partials/instructor-sidebar.blade.php	modified
resources/views/student/assignments/index.blade.php	modified
resources/views/student/challenge-quiz.blade.php	modified
resources/views/student/challenges.blade.php	modified
resources/views/student/coding-challenges.blade.php	modified
resources/views/student/modules.blade.php	modified
routes/web.php	modified
tests/Feature/Regression/Updates3ChallengeMapTest.php	added
tests/Feature/Regression/Updates3ClassChallengeAssignmentTest.php	added
tests/Feature/Regression/Updates3CodingChallengeManagerTest.php	added
tests/Feature/Regression/Updates3CodingTestRunnerTest.php	added
tests/Feature/Regression/Updates3ContentCodeGenerationTest.php	added
tests/Feature/Regression/Updates3InstructorBuilderTest.php	added
tests/Feature/Regression/Updates3InstructorChallengeVisibilityTest.php	added
tests/Feature/Regression/Updates3LessonEditorTest.php	added
tests/Feature/Regression/Updates3LevelIconsRemovedTest.php	added
tests/Feature/Regression/Updates3McqImageAndPreviewTest.php	added
tests/Feature/Regression/Updates3ModuleEndRedirectTest.php	added
tests/Feature/Regression/Updates3ModuleManagerTest.php	added
tests/Unit/LessonBlockRendererTest.php	added